

ActInf Textbook Group ~ Cohort 8 ~ Session 4

(Chapter 2) ~ 3/13/2025

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hello everyone welcome to the active inference Institute uh this is cohort 8 and we are looking at chapter two of the active influence textbook we're still in the first half of the book so we're looking at a lot more Theory a lot less practice but it's with chapter 2 that we suddenly find ourselves getting plunged into a variety of terms many of which will be familiar to folks who are interested or already familiar with probability Theory statistics basian Theory information Theory working potentially with matrices for some of our examples um and yeah statistics in general so a lot of this sometimes comes off as intimidating to those who don't necessarily have a mathematical background uh I do promise it's worth bearing through it if only to become more familiar with the terms and to try and get what you can out of some of the more sort of somewhat more common sensical maybe that's not quite the right word but examples of you know if You observe something jumping is it an apple or is it a frog right it's it's like being able to pH rephrase things in rather simpler terms helps to clarify meanwhile for those who are familiar with a lot of these terms were also introduced to uh kind of the framing for looking at things like expectations surprise how do we tie together biological inference with basian Theory what is the difference between surprisal versus basian surprise and some of these kind of key figures in the textbook really help to paint more of a picture uh I say tongue and cheek as we look at a picture um in trying to understand something like how an action perception Loop works now we've already been through the course of this chapter introduced the idea of an agent uh on or or a generative model uh there there are arguments as to whether the agent is its generative model or perhaps it's just equipped with a generative model in either case this generative model is essentially representative of whatever kind of problem or task the the agent is trying to solve or faces um and then we also have the generative process which can uh comes off as a synonym here for the environment so we there's the notion of a marov blanket which is also incredibly Central to active inference and Daniel and I were just chatting about it

just prior to this um this session and so we we notice there's nothing here directly connecting the agent and the environment other than this y here this observation that's what is emitted by the the quote unquote true hidden state of the environment denoted by X^* here and then that's used to infer the hidden state in the the mind so to speak the generative model of the agent um we notice we have Dash lines denoting sort of the Markov blankets and I think the authors themselves note this um the U for Action just as the agent takes in the why the observations the the agent emits an action um it it shouldn't be too mixed up this U uh later figures back in in a later figure and and it's it might be a little difficult to understand that it's it's actually an active state it it might be better represented here between the just as y between them U should also be between the model and the process um those are the sensory why an active U states that kind of Define like these these boundaries and also nonetheless are sort of the vicarious pipelines or passageways almost between the Markov blankets of the agent versus the environment other ideas we're introduced to are that are very important is that the way that the agent models the true state of the environment it's it's always more or less attempting to approximate something from the environment that said there's no necessity that this x is equivalent to the true state in fact there's not even a necessity that this x is structured equivalently to the true State you know if uh you know if I'm hungry and I'm tired uh there's no necessity that I think that I'm hungry and I'm tired I might be thinking in very different ways about those things um so that's just an attempt to give you a quick view of sort of what active inference is really looking at ultimately we have a partially observable environment meaning we can't actually see the real environment we're always kind of vicariously like trying to interpret it and understand it through our sensory observations perception and through action we are active in Furs we don't simply take in passive observations we or rather we don't passively take in observations we emit actions as well uh that in fact goes on to condition the environment around us which then elicits new observations so we can kind of see how the the model and the process each have their own kind of respective Logics which can work incredibly similarly uh but they also don't have to or at least they don't have to be structured the same um if I knock on wood because it makes me feel better is it actually doing anything other than kind of making me feel better sometimes I do it anyway because I know it makes me feel better because I have that mental Association it's that sort of sort of logic that I'm I'm trying to get at and so that it's just a reminder action as inference we act it's part of the inferential process right and then some other key ideas are looking at like minimizing variational free energy um this essentially what we're trying to do

or the claim of active inference as well as potentially the Basi and brain hypothesis and other related areas but but this is of course looking directly at the FP the free energy principle um we're looking at trying to uh minimize our uncertainty of course uncertainty prediction error variational free energy expected free energy surprisal basy and surprise these are all various terms that we find through the book they all do have their own nuances to them they come into play at different times um but in some sense they're very relatable uh conceptually at least but for variational free energy this is the this is kind of the the trick and it's not just a trick it plays out mathematically and we see we see it with Jan's inequality but variational free energy the whole reason why we're doing this uh compute why if the goal is to just minimize our surprise you know so that we can better understand everything why are we including this variational free energy term why do we have a variational free energy term why do we have free energy principle at all um the idea is that surprisal is unfortunately very difficult to compute um for those who have worked with let's say like deep learning and neural networks right if you have a fully connected Network there's an immense amount of information being passed uh depending on how you set up your parameters and what kind of state spaces you're working with you quick run into issues of intractability where it's something just quite literally like does not compute um similarly here uh with variational free energy the idea is that we get an approximation of surprisal uh it it plays out really well because through Jan's inequality uh which which we'll get to later probably in a couple weeks with chapter four I think is where uh where it is uh what what it allows us to do is say okay well we can't directly compute surprisal but based upon particular assumptions as well as as well as mathematical like proofs and how it plays out surprise uh can actually be upper bounded by variational free energy there's no necessity that variational free energy equates to surprise but it is a very fair approximation and we know that surprise might not be equivalent but we know that it won't be any higher than variational free energy with variational free energy by doing things like mean field approximation being able to sort of marginalize out variables from joint distributions and so on WE essentially are kind of carving things up in a way that is easier to work with them um it's found through different neurobiological studies and fact that we we do engage in approximate methods uh it would be very taxing on us if not intractable it'd be incredibly taxing on us to perfectly compute everything at any given moment uh including Trying to minimize our uncertainty and that's not even getting into the other sensory aspects such is like you know just because I can see everything in my field of vision it doesn't mean that I'm actually processing very intently every single bit of what I'm taking in we're very biased

towards things that tend to be interesting to us align with our goals in terms of pragmatic value or or things that interest us that we might learn from in terms of epistemic value so with free energy we we see that there's this process of kind of engaging in free energy minimization relative to our engagement with the uh environment all through reception changing your mind to minimize the difference Oh I thought it was 5:00 but I look at the time it's actually 6:16 I'm going to change my mind it is 6:16 or action to change the world I am hungry I'm going to eat food now I'm no longer hungry in this case this is has to do with the idea of interoception I'm reading my own signals from my body in that in that instance it's as if my own body is the environment for me right uh there's a nice neuroscientist uh Yori busak who said that the the brain doesn't know I think at least Elman Barrett said this too you know it's a bunch of neurons right so and many other things too but but it's conceptu very interesting so um I've been going for some time expected free energy we'll just touch on that real quick um I used earlier this phrase uh pragmatic value right food is in my environment I'm hungry I expect to not be hungry so I will take an action that minimizes expected free energy in order to saap my hunger by eating the food that is my action Information Gain I already mentioned that earlier that's I thought it was 5 o'l I check the time turns out I have new information I didn't really accomplish too much but I did learn something those are some kind of more everyday ways of attempting to describe how these Dynamics play out these kinds of equations are very important to be able to get some intuitive grasp of especially if you're still waiting your way through the textbook um it's great part of the the the strength of active inference that I've always appreciated is being able to apply these sort of more qualitative terms to different aspects of active inference such that you don't necessarily have to know all of the underlying maths in order to talk about active inference or do research in active inference or write about it um you know you don't necessarily need to know it in any kind of deep sophisticated sophisticated way as long as you're able to link the terms up appropriately um never a bad idea to work with a computational modeler or a a statistician or otherwise to to Really clarify your thoughts but nonetheless um very Poss there have been many interesting Papers written in active inference on phenomenology cognitive science uh social theory uh imagining ways that we could apply active inference to a variety of fields mental health um so um there's plenty in this chapter I believe it's the longest chapter in the book um there's a lot we could touch on so um is anyone interested in talking about any piece of this or anything that stuck out or any Curiosities questions that came up over the course of reading it yes Stephen please go for I apologize if there's background noise I just wanted to say that I in going over the chapter and

looking at some YouTube videos to try to understand it better I found um artam Ken's YouTube on uh the cross entropy um term or the reason why that is used separating out that um the Q value the the uh I don't know all the terms but uh where you don't have to do the complete computations you just know what the um Divergence is um it was it was what what really clicked for me was the variational um what's it called um V variational free energy variational proba um uh using the analogy of image generation from Deep learning um drawing from the the probability configuration the data structure of images that they were trained on and in the in the brain of course the data structure that were trained on is the world and and the Brain learns the data structure just from trial and error from experience so the brain figures out the variational free energy I mean it it it it it um is it's deriving or it it yeah learns the variational free energy from you know what is what are the probability data structures of the world which is why the more experience a person has the better predictions one makes uh the the the more interaction with the actual World instead of simulations of the world or imaginations of the world uh the better the predictions become I mean it just this the explanation helped me understand what why variational free energy is the the the right framing um you know uh comparing it to the brain how it grows and develops and what we're left with um I and I and I had a question I I wonder if it's okay to ask um you can tell me if this will be covered later but I've i' I always want to try to see how this is being applied in the sense of the way the nervous system is set up where there is constant sensory input and motor output but that motor output is both sent to the effectors but also sent to the internal model and the internal model or motor um a copy of the motor command is sent to a what we're going to call the internal model but the the networks that will respond to whatever the motor effectors are doing and it's comparing are the predicted sensory responses the same as the actual sensory responses right and when there's a difference then the model is updated or you know the neuron synaptic weights are changed and I'm I'm wondering how can I see that in this active inference model like in the figure that you just showed uh you know where there's the internal model and then there's the world and we're we're updating the model it didn't look you know in my mind there are parallel paths there's the motor command and the internal copy it's called the eer copy motor command and the internal motor command is creating it a it's activating a internal model of the sensory input that would be the result of that motor command which is then compared with the actual sensory input of that motor command and that's the way we're continually improving the actual motor commands so yeah so there's I will say you will definitely get more out of this textbook in terms of what it is you're describing and what you're

interested in um I just want to quickly find because there many not trying to give anyone uh you know Whiplash here but I'm just kind of jumping around a little bit um so with chapter I think it's at the end of chapter eight and I'll and I'll just hang on this figure for a bit um yeah so a few different things um and this is more like you know there will be some loose ends that kind of open up here but a few terms you might want to look for in the textbook um that's the fortunate thing about this being freely you know uh downloadable online and uh pulling up a PDF and just hitting control F to just kind of Zoom around through the textbook uh a few phrases are proprioception uh sensory attenuation um this might relate as well in terms of like trying to directly model um some of the Dynamics you're describing which are H uh this this would be a what is at once a hierarchical and mixed or hybrid model um and it doesn't necessarily have to be hybrid but all those things first proprioception um is related to to kind of like so so with inactive inference we have what are called Observation modalities it's a formal term certainly not unique to active inference but it's you know my eyes allow me to to see and so I have a visual observation modality uh and so it's like I'm taking in visual information so you can think of that as a data stream and and I have the the receptor or the sensory receptor for accepting visual stimuli or observations right um and then and then there's exteroception or exteroceptive observation modalities which would kind of be describing things that are coming from without these are very cognitive kind of Neuroscience related terms so um you know they're kind of human centered or at least organism centered but but Inception would be signals coming from within exteroceptive signals would be those coming from without and then proprioceptive would actually have to a lot more to do with kind of movement uh it could be you know in the sense of like a reflex arc for example um so from there sensory attenuation which is always something interesting to wrap one's head around um in active inference it's it comes off as a little counterintuitive but actually once I understood it it was one of the most fascinating Concepts they have and it makes sense from within the framework which is much of active inference you know given the basian lens we're always bringing some kind of Prior or what you expect to the situation what you that that includes prior on uh hidden States uh like what you infer to be going on uh but then there's also priors about observations right and so whenever you're describing this kind of aspect of having a model up for the sensory stimuli itself itself for the data stream where it's you know it's it has to figure out what the observations themselves mean um you could view that as and and then and then once it kind of figures out what they mean it it sends that further to another part of say a model to engage in further or deeper inference here this hybrid model is kind of saying at

this very fast time scale I'm taking in these observations rather rather quickly right at each step of this slower model up here a single you know this T minus one or to minus one so at one time Step at the higher level we actually have three time steps at the lower so the idea is that at different levels you can have different time scales of the inference process but all this is to say and I'll try and because I hope I I prefer to clarify more than I make ambiguous uh you know an answer to your question but um it would be something like you could model what does the incoming data mean and then from there based on what you think it means how does that fit into another inference process for making higher order decisions or making higher order inferences and given the whole partial observable Paradigm uh it is I mean yeah there's there's no necessity that the data you receive is one to one with the Real Environment um I mean this account given the you know the the the more human Centric or organism Centric robotic Centric aspects of active inference it's like what happens whenever you have a bad sensory receptor right like you still have to kind of figure out what an observation is or means and so you could imagine like in a in a in a more realistic sense I mean this could be an incredibly long complex like deep um uh model here where so many different inference processes are happening in different time scales you know layering things up then the only thing is why do any kind of modeling if in reality everything is that complicated or at least we think it is well with active inference as well as any other form of machine learning or deep learning we're ultimately just trying to come up with a good model that generally approximates uh the the phenomena of Interest or at least can can predict on it um and it needs to be not too complex otherwise it'll overfit uh and and also be computationally inefficient you know like me describing like earlier like you know if I took in every single bit of information and computed every single bit of information every bit of surprisal that there could be I mean I'm either going to hit intractability or massive exhaustion in a certain way um and and yet accuracy so I want it to be good um and I guess referring to where you you started with variational free energy that that's right I mean it would be to to approximate um so so to to create a model itself isn't just to try and be hyper accurate that's not just because of intractability or difficulty it's also because it might not be a good idea to do that as soon as there's some kind of paradigm shift in the environment right right so like if I'm in my home versus in a car I will act very differently right um if I am overfit or I do overfit my generative model to being indoors then as soon as you put me in a driver's seat I will have no idea what to do everything will be very surprising any kind of strategy I would have will be biased towards something I would do indoors it would be problematic so having approximation and

generalizability goes beyond just you know uh any kind of Cheap Trick uh uh not that anyone has said that but it just like just it's it's not so simple it's it actually helps for adaptation right so yeah if I could that was wonderful the the nervous system of course is composed of 10 to the 10th individual predictor you know models you know neurons and it's organized to selectively suppress everything but what is needed at the moment and I I see what you what you've showing is kind of giving an example of it uh on on so many different levels from The Reflex level to the cognitive no I want to be doing this now and say centering attention on one particular model yeah no that's great I'm glad this was helpful and it's you know so this model here is just jumping to chapter eight right so this is this is Way Beyond where anyone you know who has just started the textbook and they're only at chapter two now like that they wouldn't even have to consider any of these pieces of architecture yet other a couple things like we saw the Y earlier your observations and then X the hidden States but all of this about G about F functions about you know Adas and so on a matrices that hasn't quite figured yet we do have G expected free energy so it's kind of like this is you know it is a textbook so you can certainly like kind of build your way up to this and frankly I think this would be one of the most complex models that's shown throughout the textbook um yeah by the time we get to chapter nine it actually reverts to well not reverse but this is more about modelbased data analysis so this is more about kind of taking what we've learned adding on a few other forms of analysis for kind of like understanding our models or building or iterating overbuilding models and and even fitting models uh and then using them for uh you you know by fitting them we can get into kind of empirical data analysis as well as like um you know if you're working with with groups of of individuals that you're analyzing right then you can you know look at summary statistics or or other kinds of analyses per group um but yeah it's yeah it's really just getting that initial intuitive grasp figure 9.1 has always been kind of fun um you know it's this is uh a PDP model which is just a discrete State space model which you know you could you it's not dramatically there's certain kinds of conceptual overlaps this could happen with like a hot field Network or something along those lines where it's like you know you could have States just being a neurons charge being positive or or negative or or something is excitatory or or inhibitory or otherwise kind of like Reet valued um but anyway the the bigger point of this figure 9.1 as we zoom out is that we're just being aware like we have our own we the experimenters have our own idea of what the model is and what it should be so it's rather meta but I've kind of appreciated the sort of self-awareness that uh that active inference practices because at the end of the day we are making a lot of decisions anyone

working with with neur neuroimaging data or otherwise is making a lot of decisions whenever they start modeling and there are things that you know will be these kind of blaring never leave the back of your mind like sort of issues with modeling that are like oh I I need to include that but I have no idea how and I can't even you know so so we have to make certain kinds of like hard decisions sometimes when it comes to modeling and to bring it back back um that would be sort of the case for organisms too right like like all this computational modeling we're doing based on textbooks like this and reinforcement learning and so on and apologies if I'm if I'm getting too abstract or philosophical but it's like similarly you could think of the brain as like well you know if I start a new job uh you know place of employment and I'm trying to figure out like you know how does this work what are my social dynamics with my co-workers you know what is kind of our our degree of professionalism and and um you know how do we how do we sort out doing meetings and who is the best person to talk to about this topic and all kinds of things I'm going to have to start kind of at least implicitly modeling how this goes because I'm going to presumably be engaging with my job like you know X number of hours per week or you know scheduled meetings or otherwise um so so any any scenario would come to whether it be trying to model you know myself properly engaging in my position of employment or it' be us trying to model how a mouse in a teammate's behavioral task you know functions and how it is that they infer where the reward is on the left side or the right side of the teamas and what are the steps that they take to get there and how do they decide that or determine that based on their uh sensory observation I mean it's it's just a a lot of modeling to do and and it's almost always approximate life moves quickly um I mentioned sensory attenuation too earlier that's kind of a fun one but I don't want to introduce too much too quickly let's see if this is a good place to pick up on that the implications that we need to be able to attend away from proprioceptive data by turning down their gain this gain is given by the Precision or inverse variance with which these data are predicted to attenuate this we need descending motor tracks to predict not just the data but the Precision of or confidence placed in those data decreasing this prision to initiate movement while this for some will come off as very complex passage and it it really is but but this is chapter 5 so there's a good deal that you'll have learned by the time you reach this area sensory attenuation is interesting in part because like as a as sort of a learning hortoc for active inference I think because it what it what it revolves around is this sort of basian Paradigm that is we bring our own beliefs to the situation and a lot of things have to do with expectations right so there's a kind of strange philosophical conundrum that can come up when considering

that well if you expect that right now you are sitting in place you might have a goal to you know I might have a goal to stand up but currently I expect myself to be sitting you that's that's what I'm observing right now I notice I'm still sitting down so how if it's the case that we engage in like bayes and inference then how is it that I ever get up from the chair at all if my priors or expectations are always reinforcing um you know what I'm doing or what I expect and that's influencing my action so sensory attenuation is the sense that in everyday terms very everyday terms because I notice the time and I'm kind of droning a little bit about this but the idea would be that it's almost as if your sensory receptors have to not pause but heavily turn down their own gain like in order to actually finally get up out of the chair I have to it's almost as if I'm forgetting that I'm in the chair I need to move in order to make something happen and so it's a rather interesting framing for looking at how we actually accomplish goals while reconciling this idea of you know the fact that we're we're always inferring um give me just one second because I realized the chat's been very um let see there was a question about this is backtracking but let's uh because it'd be good to get ourselves reentered in chapter two been taking us on a journey um so expected free energy we'll go to that equation I will look a little bit more at some of these terms so like here there's a question about pragmatic value so what is this the expectation of the natural log of the conditional probability distribution y with a Tilda meaning it's a sequence or it can be a sequence uh one plus elements in the sequence conditioned on C so I don't know if it was nice of the authors or not to so quickly introduce us to G whenever we're still missing some things like what is c um but that's okay so C uh here I mentioned earlier the the π the pomdp model that we were looking at which is for discrete State spaces I was zooming in on those figures and chapters 8 and chapter n um to C is a term denoting your priors over observations so this is kind of like saying if we just reduce to this it's kind of like what is the probability of this sequence of observations conditioned on the observations I expect and so pragmatic value is the idea is very much related or almost the way it's used is almost identical to the idea of reward seeking has so so it's not about seeking out information for its own sake to change your mind it's about directly going to an award uh reward if I expect to not be hungry then if I'm observing that I'm in fact not hungry or I'm I'm you know then then this probability is going to be very high right like my model is accurate in terms of achieving pragmatic value on the hunger end of things okay so uh from the text that said the SE is the preferences so basically this probability is the probability that you see the observations that you would want to see right because it means you have the string of observations in in the world that is your preferences which means you

get the probab means you get the observations that you want to see and it's the probability of that couple questions though uh why the logorithm and also um it it has to do with the probability that you see what you want to see but it doesn't say anything about how much you want to see what you want to see typically pragmatic value isn't about just How likely you are to get what you want but how much you actually want what want yeah just a few a few quick answers there the shape of C is a distribution over the observations it does exactly tell you how much you prefer different observations like you could have um 131 or 121 that's a sharper preference in the second case why is it a log um there are several convenient properties of logs for probability which is why likelihood distributions are often framed in terms of log likelihoods for example yeah EX exactly like those are the different sharpness but log um allow properties like $\log$ of $a \cdot b$ is $\log$ of a plus $\log$ of B so if you have two independent probabilities you can separate them so those are basic log properties and also for computation when you're dealing with low probabilities it's easier to store a negative log as a large number like 1 billion rather than a tiny number and it has properties that are important for statistics mapping between the zero to one interval so there's a ton of reasons why logs are used in general in statistics all of those are in play and then the preferences are a distribution over observations and they can express the strength and the directions of preferences yeah thanks Dan oh let's uh have another quick look at that exactly so it's kind of like just as Daniel mentioned like if C is sort of a distribution over all of your preferred or or expected observations and then you get uh a sequence of observations and you could view this you know in the PDP in the discrete State space setting you could view this as like oh um you know there's there are three different possible observations uh I prefer the first one with full probability 1.0 the other two is zero which means I completely do not want observation index two or three I just want the first possibility out of the three um so um like uh I'm I'm I'm hungry I'm completely famished or I am fully sated I prefer being fully sated right so the sequence of observations could be you know an observation of being fully stated not the others and then once again being fully stated not the others but then for maybe a third time step now I'm hungry right so there would be you know of course I'm just kind of making this up as we go but you know this so that would mean this probability would not be exactly one uh if I were sated for all three of those time steps then it would be one right it be that be exactly what I expect and then it's called expected free energy and that's some folks are like well why do we have a variational and then we have an expected the and why is it called expected and so with expected free energy this is free energy of the future so to speak and it

does kind of have its own set of equations to it but we'll notice that it's being scored for each action or policy that could be taken so a policy would be a whole sequence of actions into the future and so whereas f is pretty let's say reactive in a sense in that it takes in an observation Computing free energy to compute your posterior overstates meaning I took in this observation what is the free energy that helps me Reach the the sort of optimal approximate posterior Over States if my stomach growls best inference would be I'm hungry probably expected free energy will be oh there's a series of things I could do in the sense of actions and sequences of actions I could take so given I could take an action associated with a sequence like I could get up and go to the store to get food um by the end of that sequence at least uh hopefully at some point involves me eating the food and no longer being hungry right so to think about it computationally simply that g is just scoring each of the kind of actions or policy sequences of actions that you could take we want to minimize G as our optimization function these two terms is kind of all you need um to look at these others two but but for now we'll just focus on this um this is kind of like saying by doing a set of actions you know what will be its pragmatic value how closely does it match my expected observations my preferences what I want which from a you know a more General perspective C could very be well be what defines your constraints for maintaining homeostasis so it so C could also be oh my blood oxygen level is you know at a proper balance or or uh hunger I I keep going to that one that's a pretty good example of achieving homeostasis to make sure your uh caloric balances in place and then Information Gain is kind of like it's it's interesting because as opposed to variational free energy where K Divergence is you know it's it's kind of we want to minimize it here we're actually wanting to to maximize it because both of these terms are negative right if we want to minimize G then we actually want to maximize Information Gain and pragmatic value so with Information Gain you can look at this as what is what do I believe or what will I believe given the observations I think I'll come across when engaging this set of actions versus what do I believe is usually the case when I do that set of actions so really the only when when looking at the taale Divergence here the only thing that separates these and makes it a you know a non-zero Divergence by not equating the same two things or or or putting JS opposing the same two things is is the why so so we're looking at a kind of scenario where it's like I'm seeking out doing things that bring about observations that actually change my mind and ultimately because of the particular terms here it's kind of like saying I want to find I want to do things that change my mind about the association between what I think is going on and my actions and the the more intuitive way of saying that is simply I'm going to

do things that get help me get a better understanding between how I think the world Works based on what I do right so imagine someone who opens up a textbook to learn something right and then you you could start learning about you know oh I could apply active inference or I could think about active there are all kinds of ways that one could attempt to use active inference whether it be for you know brain computer interfaces or it be used for you know uh agent-based modeling in the social sciences looking at you know social media interactivity but you know patterns in in networks um looking at neuronal Dynamics so so it's that both of these are sort of like directives or incentives that we have and then sort of the classic bit with active inference that's repeatedly stated in a lot of the literature when comparing it to reinforcement learning is that it's these terms that are directly implicated whenever people consider this sort of explore exploit uh tradeoff so this this is a point I've made personally I uh you know I did a workshop at at upen at a conference last summer and and you know it was interesting to a lot of people who weren't familiar with AC of inference yet but did reinforcement learning because when with reinforcement learning obviously not always the case but there are at least times where people resort to defining some kind of term for what what what is e exploratory Behavior you know which is sometimes it's as simple as like a Epsilon greedy meaning uh you essentially saying like oh the agent should always seek the reward uh point you know nine times out of 10 and then point one like or 10% of the time just do something random and then that way you can avoid getting stuck in a local Minima um but here we notice like oh as long as you have the equation for G then exploitation and exploration are both scored on their own individually there's no hard coded 10% of the time you do something random right so if expected free energy computation holds and we find that at a variety of levels of you know life whether it be neuronally or for an entire person or potentially even an entire group of people depending on what level of the nestedness of of markup blankets you're looking I mean this is a way of saying oh we don't need to hardcode a particular value to say when you should do something random in fact this isn't even random right because you're actually making direct references to particular sequences of observation you expect or States you expect or actions you take so that's sort of the that's a this is kind of a major strength of active inference and for myself is someone who who's done a lot of kind of reinforcement related work in the past this is this is this was one piece of kind of why I headed in this direction because I just it seems so arbitrary to do Epsilon greedy type stuff um so is there any anything in the few minutes we have left any final thoughts any final questions someone wants to pick up on reminder to everyone that we have a textbook Koda this was a

nice video that Daniel and Ali did some time ago they they did a sequence of book streams where they did rather uh much more in depth sort of overviews of each chapter sorry about that and uh and it's kind of a nice way within less than 30 minutes to go even deeper than I did during the course of this particular session so with a lot of highlights on particular things that are worth putting your attention towards uh if you if you want to get the most out of each chapter and then for the sake of this cohort since we are out of time uh so chapter three will be next that'll instead be the high road rather than the low road to active inference so here we started with uh you know we started in chapter two with with Basi and mechanics took that route and then chapter 3 will be about starting with the high road starting from the free energy principle starting with Notions of self-organization surprise minimization see much more detail on marov blankets the idea of self- evidencing including that trick about the sensory attenuation I mentioned earlier um predictive processing so we'll we'll reach there and then chapter after that chapter 4 we'll start to bring it back together significantly more about message passing looking at actual like microc cortical circuits uh in terms of like neuronal message passing uh in cortical layers um so for anyone interested in neurobiology um that's a particularly fascinating one um I always have a good time with it personally so um all right everyone um thanks so much there thanks Andrew by everyone